Anudeep Cherukupalli
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EDUCATION

Arizona State University **May 2026**
Bachelor of Science in Computer Science (GPA: 3.6/4.00) | New American University Scholarship | Dean's List

TECHNICAL SKILLS & CERTIFICATIONS

Languages: Python, Java, TypeScript, SQL
AI Systems: LLMs, Agentic AI, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, LLM Evaluation, Prompt Engineering, LangGraph, LangChain
Backend: FastAPI, REST APIs, Microservices, SDK Design, Distributed Architecture, Event-Driven Systems
Cloud: AWS, Docker, Kubernetes, Helm, CI/CD
Data: PostgreSQL, SQLite, MongoDB, Redis, Kafka, pgvector, Pinecone
Infrastructure: Terraform, GitHub Actions
Observability: OpenTelemetry, Prometheus, Logging, LLM Latency & Cost Monitoring
Certification: Claude Certified Architect – Foundations (Anthropic, 2026)

EXPERIENCE

AI Software Engineering Intern | Qbotica **May 2026 – Present**

- Architected **agentic AI workflows** using **LLMs, LangChain, and LangGraph** to automate multi-step enterprise operations across logistics systems, email, APIs, and internal applications with human-in-the-loop escalation
- Engineered an **AI-powered logistics automation platform** integrating enterprise TMS/ERP systems through an API-first architecture with **UiPath RPA and computer-use agents** as fallback layers for legacy workflows
- Developed a **universal data extraction and normalization pipeline** that transforms unstructured emails, documents, shipment requests, and TMS records into standardized schemas consumed by downstream AI agents and backend services
- Built **multi-agent orchestration workflows** for request triage, route and capacity analysis, pricing, load optimization, validation, and automated communication, enabling end-to-end execution of complex logistics workflows

AI/Backend Software Engineering Intern | Grey-Box (Project CyberCafe) **Sept 2025 – May 2026**

- Designed and maintained **8+ RESTful API integrations** across multiple microservices, enabling consistent data flow across distributed systems supporting AI workflows
- Optimized **12+ data pipelines** processing **300K+ records daily**, resolving **15+ synchronization issues** and reducing recurring production issues through improved validation and monitoring
- Deployed services across **3 AWS environments** with **10+ validation checkpoints**, maintaining reliable system performance across distributed services

Backend Software Engineering Intern | SVSRco **May 2025 – Aug 2025**

- Developed scalable backend API services with **20+ endpoints** handling **3K+ daily requests** at **¡200ms response time**, enabling high-throughput processing
- Engineered cross-system data integrations across **6 platforms**, handling **120K+ transactions** with consistent data flow between backend services
- Troubleshot **25+ production incidents** using logging and monitoring tools, reducing recurring issues and improving mean time to resolution

Instructional Aide | Arizona State University **Aug 2023 – Dec 2023**

- Guided **120+ students across 4 lab sections** in algorithmic problem-solving and system design through **30+ office hours**, reinforcing core software engineering principles
- Conducted **200+ code reviews**, improving code structure, debugging practices, and adherence to development standards

PROJECTS

Multi-Agent Intelligence Engine | *LangGraph, Python, Redis, PostgreSQL, OpenTelemetry*

- Orchestrated a multi-agent system with **5 specialized agents** using LangGraph with policy-based routing, enabling coordinated task execution across research, risk scoring, compliance, reporting, and human-review workflows
- Implemented state persistence and observability using Redis-ready runtime snapshots, SQLite/PostgreSQL checkpointing, and OpenTelemetry-based telemetry, ensuring traceability and human-in-the-loop escalation

Enterprise RAG + Evaluation Harness | *Python, Pinecone, pgvector, RAGAS, GitHub Actions*

- Engineered a hybrid retrieval pipeline over **1,000+ documents** combining semantic chunking, BM25 retrieval, semantic search, and deterministic re-ranking for healthcare compliance question answering
- Developed an automated evaluation framework with RAGAS-compatible exports, synthetic QA generation, and quality measurement across **8 metrics**, including faithfulness, answer relevancy, context precision, recall@3, safety, latency, tokens, and cost

WorkflowGuard – AI Workflow Evaluation Platform | *Python, FastAPI, Next.js, PostgreSQL, LLM APIs*

- Architected an **AI-powered workflow verification platform** for BPMN, JSON, and n8n workflows, combining graph-based static analysis with LLM-assisted prompt alignment, reliability, security, and maintainability evaluation
- Engineered an **automated workflow QA engine** with AI-assisted test generation, safe simulation, failure injection, coverage analysis, cost forecasting, and controlled repair generation with human-reviewed versioning
- Built **CI/CD quality gates** enforcing structural, security, reliability, prompt-alignment, test-coverage, and cost thresholds, with a FastAPI/Postgr backend and Next.js dashboard for evaluation, audit history, and reporting